A Written de Sitter–Unruh Modified-Inertia Action (v13)

The passive-frame constraint structure closes (a machine-verified Dirac system with zero propagating frame modes), the nonlocal operator is rigorously defined by a positive Herglotz–Nevanlinna measure, the theory is causal, ghost-free, Cassini-safe and nonlinearly stable, gravitational lensing is resolved by a ghost-free disformal photon metric, the one-loop de Sitter radiative structure is computed at the divergence level (a_0 unrenormalized; no aether kinetic term generated), and the two-loop transverse-aether channel is closed at divergence level in both the matter and graviton sectors (a_0 additive non-renormalization all-orders-exact)

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Abstract

The de Sitter–Unruh reading of the MOND acceleration scale, $a_0 = cH\_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36\times10^{-11}$ m s^{-2}, posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. This paper writes the explicit nonlocal modified-inertia action and carries its covariant analysis to a definite boundary, reasoning throughout from the framework's own premises (a passive frame, with the MOND in the matter sector) rather than the standard aether/MOND lens. Results, each with reproducible verification (both a_0 footings): **(1)** the **worldline/matter sector is machine-verified** ($\nu(y)=\sqrt{(1+1/y)}$ to 3×10^{-13}, Newton, deep-MOND, baryonic Tully–Fisher, the dS-Unruh-forced $\sqrt{2}$-DC-weight external-field kernel, exact ghost-freedom). **(2)** The modified-**inertia** realization **evades the Cassini quadrupole** by ~7 orders (deep-Newtonian $\nu-1\approx7\times10^{-7}$), where the modified-gravity realization fails at +6 to +14σ. **(3)** Promoting the frame field covariantly, **MOND does not force modified gravity** — the frame-equation source is l=0 (soaked by the unit-timelike multiplier) and the matter stress tensor's *principal* (UV) part is likewise an l=0 isotropic source, so the l=2 Bianchi shear lock that forces AeST's Cassini quadrupole is not populated at that order; computing the *full* metric stress tensor T_μν and its subleading multipoles is a named open step. **(4)** The mixed matter–aether **two-point propagator is causal and ghost-free in closed form** (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol = the GR light-cone). **(5, correcting v2)** The Einstein-aether **strong-coupling wall does *not* wall the framework**: that wall is a property of a *propagating* aether mode whose kinetic norm vanishes, but the framework's frame is **passive** (its own theory: an algebraic constitutive law, an SME background, the cosmic rest frame), with zero propagating aether modes — so the wall's object is undefined for it. The wall binds only the rejected AeST/khronon *modified-gravity* limb. **(6)** Gravitational **lensing — a disformal photon metric (universal, Cassini-safe, no double-counting)**: a single-metric audit shows that modified inertia, calibrated to the RAR with the baryonic field, makes light *under*-lens by ν, and that enhancing the one metric to fix it **double-counts** the dynamics (forcing pure modified gravity, which reinstates the Cassini bill). The resolution is a **disformal photon metric** $\tilde{g}_{\mu\nu}=g_{\mu\nu}+B\,u_\mu u_\nu$ *on which* light *propagates while matter keeps modified inertia in g. Only the disformal $u_\mu u_\nu$ term bends light (a conformal factor is null-inert), and fixing B by the same ν the RAR uses makes lensing* ***universal*** *(one rule, all galaxies — not per-galaxy tuning),* ***double-count-free*** *by construction, and* ***Cassini-safe*** *($B\sim(\nu-1)\longrightarrow a_0/2a\longrightarrow0$ at high a, giving a PPN shift $|\Delta\gamma|\sim10^{-13}$ at the Sun, ≪ the 2.3×10^{-5} bound; the passive frame carries no dynamical-aether bill). The two-metric coupling is* ***ghost-free and causal, nonlinearly*** *($\tilde{g}$ Lorentzian, photons subluminal*

vs g, no new dof; the coupled Ostrogradsky check is done — the disformal Lagrangian is first-order in every dynamical field given §4 passivity), the one hinge being the passive frame. This supersedes the fine-tuned ghost-condensate of earlier drafts. New in v4, addressing three formal completeness questions: ***(7)*** *the* ***passive-frame constraint structure is a machine-verified closed Dirac system****: the full curved-spacetime Dirac analysis terminates at the secondary level (the unit-norm sector is genuinely second-class — its 2×2 Dirac block has determinant $4(u\cdot u)^2\longrightarrow 4$ on-shell — with no tertiary tower), leaving* ***zero propagating frame degrees of freedom*** *(2 total: the GR graviton). The crux — that u sits* inside □u and might thereby acquire a kinetic term — resolves negatively: the exact quadratic symbol at every order is $S_n=(-1)^n\ k^2\perp\ k_0$^(2n), whose only root is $k_0=0$ independent of spatial k, i.e. a transport-along-u ODE with **no wave-cone and zero group velocity** — the matter coupling does *not* promote u to a dynamical field. **(8)** The nonlocal operator **K(□_u) is rigorously defined** by the Borel functional calculus of the (essentially self-adjoint) directional operator □*u, with a canonical* ***Herglotz–Nevanlinna spectral representation*** *carrying a* unique positive Borel measure *(closed-form density, Källén–Lehmann-positive); K is bounded (‖K‖≤1, operator-monotone) on all of L^2, causal-retarded, and its non-entire branch cut is the physical de Sitter–Unruh emission continuum, not a ghost.* ***Nonlinear*** *classical stability follows from the same no-wave-cone symbol to all orders together with the matter sector* not *modifying the graviton principal symbol — because K⟶1 in the UV its leading contribution is a non-derivative source, so hyperbolicity is inherited from GR (computed, not assumed);* ***(9, new in v11)*** *the* ***one-loop de Sitter radiative structure is computed at the divergence level****: the complete matter-loop counterterm list is assembled exactly (Gilkey a_1/a_2, made possible because W=u^μK(□_u/a_0^2)u_μ is a local endomorphism, under a stated ρ_m=$m^2\varphi^2$ proxy);* ***a_0 is not renormalized*** *— no z^0 tadpole at the exact-measure level, with a new sum rule ∫dμ(t)/|t| = K(∞)−K(0) = 1 (unit resolvent weight, verified to 10^{-12}); the linear frame vertex on dS vanishes at* every *resolvent order (a theorem, via the geodesy identity $u\cdot(u\cdot\nabla)^nV=(u\cdot\nabla)^n(u\cdot V)$);* ***no transverse $(\nabla u)^2$ aether kinetic term is generated*** *— the one dangerous channel, graviton-frame mixing through dS curvature (R*{μανβ}$u^{\alpha u}$β = $-H^2P\perp$ exactly, so curvature *does* make transverse structure), closes because commutator insertions are algebraic ($k_0^2\longrightarrow H^2$, never $k\perp^2$): all characteristic roots stay k-independent, no wave cone forms, and the TT-graviton×frame vertex is exactly zero (CAS, n=1,2); what *is* generated is reported with coefficients (a vacuum-energy renormalization of the *source* ρ_m through the framework's own form factor, a longitudinal $O(\delta u^4)$ W^2 operator, and a curvature-cross term suppressed by $2H^2/m^2\sim10^{-83}$); dressed Källén–Lehmann positivity and KMS detailed balance survive the real one-loop convolution of the exact cut density (both footings; the positivity detector's measured blind spot is stated). **Open:** the T_μν/disformal ρ_m variant (argued, not computed), one-loop *finite* nonlocal parts on dS, all-n extension of the TT-vertex zero, pure-graviton diagrams beyond selection-rule coverage, and anything two-loop. **(10, new in v12)** the **finite parts and the two-loop matter sector**: the one-loop *finite* effective potential is **bounded below** (the exact Herglotz norm ‖K‖≤1 with s=−1 confines the effective mass² to (0,m²], no runaway) and the de Sitter IR is **regulated by the friction gap** (□*u gapped at $-9H^2/4$ gives every loop field an effective mass ≥3H/2, killing the usual dS secular-growth problem);* ***a_0's additive non-renormalization is now all-orders exact*** *(the shift symmetry T⟶T+const with ∂*c u_μ=0 verified for the *full nonlinear* frame, together with the unit-timelike constraint collapsing any surviving z-independent frame functional to a pure cosmological constant — so β*{a_0}^{additive}=0 to every loop order); and the* ***two-loop matter-sector transverse aether kinetic term is not generated*** *— a naive matter-bubble estimate that appeared to produce one is an artifact (it mis-assigned the K form factor to the internal loop momentum and used a δu-φ-φ bubble vertex that vanishes identically by the F1 geodesy theorem), and the correct all-orders statement is structural: δu enters only through K(*□u)⟶K($-q_0^2$) or longitudinal (u·∇)⟶q_0 factors, so the transverse coefficient at $q_0=0$ vanishes by K(0)=0 — the same fact that forbids the a_0 tadpole. The **graviton-loop sector is now also closed at the divergence level (v13)**: a complete census of the two-graviton-loop topologies finds *no* transverse aether cone — the graviton's transverse momentum q⊥ *is* ***rationed*** *(the frame's external-momentum power is zero to all n, while a* $k\perp^2|\delta u\perp|^2$ *cone would need two, i.e. a different ≥4-frame-leg topology),* *the TT-graviton×δu⊥* vertex vanishes (CAS n=1,2,3, both polarizations, plus an all-n symbol induction and skeptic-banked n=6), and the constrained h_{0i} shift sector stays instantaneous (no q_0-pole, so no propagating mode). So **the passive frame (0 dof) survives two loops at the divergence level in both the matter and graviton sectors**. **Still open:** one-loop and two-loop *finite* parts, genuinely higher loops, wiring the exact seagull tensor into the loop integral, constraint-survival under loops, and the disformal ρ_m variant. The MOND sign remains a **postulate** and a_0′s value is underived. An effective one-scale completion carried to a sharp, named boundary — **not a finished theory and not a theory of everything.**

1. Setting and scope

The framework reads a_0 as the acceleration at which a body's inertial response to the de Sitter vacuum departs from Newtonian, with the Deser–Levin temperature T(a) = (ℏ/2πk_Bc)√(a² + (cH_Λ)²) a horizon floor and the interpolation g_obs = √(g_bar² + g_bar a_0), ν(y)=√(1+1/y). The earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; this is a one-scale effective-theory construction with named inputs, each labeled **derived**, **postulated**, or **open**. Crucially — and this is the correction to v2 — the framework is *modified inertia*: the MOND lives in the **matter** sector, and the frame field u^μ is a **passive**, horizon-anchored reference (the framework's own paper: an *algebraic constitutive law* g_bar=G(a) with no frame equation of motion; an SME gravity-sector *background*; the Cassini-constrained cosmic rest frame). It is **not** AeST/Einstein-aether, where a dynamical aether *carries* the MOND — the theory that fails Cassini.

2. The action

With signature (−+++), S[x,g,u] = S_EH[g] + S_u[g,u] + S_matter[x,g,u]: host GR unmodified; a unit-timelike frame constraint S_u = −∫√−g (λ/2)(u^μu_μ+1) (no aether kinetic term); and the modified-inertia content in the matter kinetic sector,

$$S_{\text{matter}} = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,\rho_m\left[\,s\,u^\mu\,K(\Box_u/a_0^2)\,u_\mu\,\right],\quad K(z)=\frac{\sqrt{1+4z}-1}{2\sqrt{z}},\quad \Box_u f = u^a\nabla_a(u^b\nabla_b f).$$

K is non-polynomial with a branch cut and a single healthy pole (residue +1). The equivalent Galley doubled-worldline form uses the conservative gradient $G(a)^{\mu=a}$μμ_fw(|a|/a_0), μ_fw the exact inverse of ν(y)=√(1+1/y); the external-field kernel θ(y)=θ_0/(1+(θ_0−1)y²), θ_0=√2, has its shape forced by the de Sitter Wightman function. Here **s=−1 is the MOND-sign postulate**, a_0=cH_Λ/Z=9.36×10^{-11} (canonical), 1.13×10^{-10} (alternate footing).

3. Verified: the worldline sector and the Cassini evasion

Machine-checked, both footings: the circular-orbit reduction reproduces ν(y)=√(1+1/y) to 3.35×10^{-13}; Newton, deep-MOND, and baryonic Tully–Fisher v^4=GM a_0 (exact); the form-factor↔RAR identity g_obs=√(g_bar²+g_bar a_0); the √2-DC-weight external-field kernel; and ghost-freedom (single pole +1 vs the Ostrogradsky ghost of the local truncation). **Cassini:** the anisotropic galactic external field (a_ext=2.29 a_0) enters only Saturn's inertial response, deep-Newton-suppressed by ν−1≈7×10^{-7}; the l=2 quadrupole is ~7.4×10^{-34} s^{-2} — ~7 orders below the 5.2×10^{-27} ceiling. The modified-gravity realization instead fails at +6 to +14σ. This is Milgrom's [2009] statement that the inner-solar-system quadrupole is an AQUAL/QUMOND effect absent in modified inertia, realized explicitly.

4. The covariant completion, on the framework's own terms

Because the frame is passive and the MOND is in the matter sector, the covariant question is *not* whether a dynamical aether is strongly coupled — it is whether the passive-frame + nonlocal-K matter system is well-posed. Six results:

(a) MOND does not force modified gravity. Varying S_matter, the source it adds to u^μ's field equation is $J^\mu=-\rho_m s k u^\mu$: l=0, parallel to u^μ, soaked by the multiplier λ. It never populates the l=2 traceless-shear channel whose divergence is the Bianchi lock that forces AeST's Cassini quadrupole. So the covariant lift does not collapse to modified gravity; a passive frame carries at most $Q_2{}^u \lesssim 7\times10^{-35}\ s^{-2}$ over an open, ghost-free region. This is established at the *frame-equation* level (the δS_matter/δu source) and, for the metric, at *principal-symbol* order: since K⟶1 in the UV the leading matter stress tensor $T_{\mu\nu}=-(2/\sqrt{-g})\delta S_{matter}/\delta g$ is a non-derivative, l=0 isotropic source (`principal_symbol_blockdiag.py`), which cannot populate the l=2 shear. Computing the *full* closed-form nonlocal $T_{\mu\nu}$ and machine-verifying $\nabla^\mu T_{\mu\nu}=0$ on-shell — the direct Bianchi-lock object at subleading order — is the one named open metric-variation step (v4/v5 varied the action w.r.t. u and λ, not yet the metric); it is a finite classical calculation, not walled.

(b) The mixed two-point propagator is causal and ghost-free in closed form. The dynamical-u variation resums to $Q(z)=K+2zK'=2\sqrt{z}/\sqrt{(1+4z)}$; the retarded memory kernel has strictly causal support (an advanced control correctly flips it), Titchmarsh UHP-analyticity holds, and Källén–Lehmann spectral positivity $\rho(s)\geq0$ across the whole cut is verified in closed form (a control correctly flags a known Ostrogradsky ghost as $\rho<0$). The branch cut is a healthy radiation continuum, not a ghost.

(c) The principal symbol is the GR light-cone, $D(w)/w = 1 + i\ a_0/2w - a_0^2/8w^2 + \ldots$, the nonlocal correction $O(a_0/w)$ UV-subdominant and IR-gapped — hyperbolic, no secular runaway. So the passive-frame theory is **principal-symbol-well-posed**.

(d) The Einstein-aether strong-coupling wall is mis-applied (correcting v2). That wall is, by construction, a property of a *propagating* aether/khronon mode whose canonical kinetic norm vanishes ($M_{sc}\sim\sqrt{N}\,M_{Pl}$; the mode speeds carry the vanishing norm in the denominator) [Blas–Pujolàs–Sibiryakov 2009; Gong et al. 2018]. The framework's frame is passive — zero such modes — so the wall's central object is *undefined*, not violated. It binds only the rejected AeST/khronon modified-gravity limb (which fails Cassini at +6–14σ). v2′s "the covariant route is walled by strong coupling" applied that dynamical-aether wall to the passive-frame realization and is hereby corrected.

(e) The constraint structure closes as a machine-verified Dirac system (v4). The full Dirac–Bergmann analysis of the passive-frame + matter system in *curved* spacetime terminates cleanly. The primaries are $p_\lambda\approx0$ together with a *degenerate* Legendre map for the u-momenta (the velocity Hessian is proportional to the spatial k^2, so there is no invertible hyperbolic kinetic operator — the precise sense in which u carries no Cauchy data); the secondary is the constraint χ_1: $u\cdot u+1=0$; and the u-equation of motion *algebraically* fixes the multiplier $\lambda=-\rho s$ with **no tertiary tower**. The unit-norm pair $(\chi_1,\ u\cdot\pi)$ is genuinely **second-class** — its Dirac block is $[[0,\ 2(u\cdot u)],[-2(u\cdot u),0]]$ with determinant $4(u\cdot u)^2=4$ on-shell (machine-verified carrying the full metric $g_{\mu\nu}$) — so a Dirac bracket exists and the frame-sector degree-of-freedom count is ½[10−0−10]=**0**; the only propagating modes are GR's two graviton polarizations. The crux that u sits *inside* $\Box_u$ is resolved honestly: the exact khronon variation gives a *nonzero* transverse quadratic form (the naive "identically zero, purely longitudinal" reading is false), but that form is $S_n=(-1)^n\ k^2{\perp}\ k_0{}^{(2n)}$, whose only nonzero-$k\perp$ root is $k_0=0$ (a double root independent of spatial k) — a transport-along-u ODE with no wave-cone and zero group velocity, **not** a spatially propagating aether kinetic term. Two honest relabelings surfaced in verification, neither changing the count: hypersurface-orthogonality (twist=0) is the *khronon-frame postulate*, not an algorithm-generated secondary constraint on a generic unit 4-vector; and $\pi_\mu\approx0$ is a degenerate Legendre map, not a literal vanishing. The construction is thus mathematically complete **given the khronon-frame premise**.

(f) The nonlocal operator $K(\Box_u)$ is rigorously defined (v4). On a globally hyperbolic passive-u background the directional operator $\Box_u=(u\cdot\nabla)^2$ is essentially self-adjoint, so $K(\Box_u)=\int K(\lambda)dE(\lambda)$ is fixed by the Borel functional calculus. K(z) is a genuine Herglotz–Nevanlinna function of $z=\Box_u/a_0^2$ — proven analytically via the uniformizer $w=\sqrt{(1+4z)}$ ($K=\sqrt{((w-1)/(w+1))}$ is a composition of upper-half-plane maps) and numerically (min Im K over the upper half-plane = $+4.8\times10^{-6}$, zero violations). The Herglotz–Nevanlinna representation theorem then supplies a *unique positive Borel measure* with closed-form density ($\rho_A=(1-\sqrt{(1-4|t|)})/(2\pi\sqrt{|t|})$ on $-\frac{1}{4}<t<0$, $\rho_B=1/(2\pi\sqrt{|t|})$ on $t<-\frac{1}{4}$), everywhere ≥0 (Källén–Lehmann), with $\int d\mu/(1+t^2)$ finite; the representation reconstructs K to $<10^{-11}$. $K(\Box_u)$ is bounded ($\|K\|\leq1$, operator-monotone/ Loewner), defined on all of L^2, and causal-retarded. The non-entire branch cut is the physical de Sitter–Unruh emission-threshold continuum, not an obstruction. One relabeling improves the physical reading: $K(\Box_u)$ is self-adjoint on the static $z>0$ sector, while on the dynamical $z<0$ (oscillatory) sector the spectrum lies on the cut and $K(\Box_u)$ is **normal** and causal-retarded (its positive anti-Hermitian part is the Källén–Lehmann-guaranteed dissipative response), not globally self-adjoint.

What is open. Nonlinear classical stability now follows from (e): the exact symbol $S_n=(-1)^n k^2{\perp}\, k_0\text{\^{}}(2n)$ holds at every order, background gradients and curvature add strictly lower-derivative pieces that cannot flip the cone, so there is no nonlinear ghost or runaway and hyperbolicity is inherited from GR — this last step now *computed*, not assumed: because $K\longrightarrow1$ in the UV the matter sector's principal part is non-derivative, so its stress tensor cannot enter the graviton kinetic symbol (`principal_symbol_blockdiag.py`; $K\longrightarrow0$ in the IR, both bounded). **Radiative stability is now computed at the one-loop divergence level (v11; `mi_oneloop_desitter.py` + the three `oneloop_lane*.py` companions, each adversarially verified by independent re-derivation).** The framework's own Herglotz measure converts the nonlocal vertex into a positive superposition of local massive resolvents, so the complete 4D one-loop matter-loop divergence list on dS follows from the textbook Seeley–DeWitt a_1/a_2 coefficients — exact (not leading-order) because $W=u^\mu K\, u_\mu$ is a local endomorphism, under a stated $\rho_m=m^2\varphi^2$ proxy for the quantum matter density. Computed: **a_0 is neither additively nor multiplicatively renormalized** (no z^0 tadpole at the exact-measure level; the new sum rule $\int d\mu/|t|=1$ leaves no spare spectral weight to feed one); the **linear δu vertex on dS vanishes at every resolvent order** (theorem: $u\cdot(u\cdot\nabla)^nV=(u\cdot\nabla)^n(u\cdot V)$, from geodesy and metric compatibility); **no transverse $(\nabla u)^2$ counterterm appears** — every quadratic δu structure, including the cross-chain and second-order-constraint pieces a first scan missed (completed by the verifier), is longitudinal-only. The dangerous channel — **graviton-frame mixing through curvature**, which flat selection rules cannot see — is real ($R_{\mu\alpha\nu\beta}u^{\alpha u}\beta=-H^2P\perp$, exact) but closes: commutator insertions are algebraic, converting k_0^2 into k-free H^2 mass-type dressings and never producing $k\perp^2$; all characteristic roots are k-independent, the khronon sector keeps its overall $k\perp^2$ factor on dS, cone-forming k^3/k^4 terms are identically absent, and the frame-linear vertex with the *propagating* (TT) graviton is exactly zero (CAS, n=1,2, both polarizations — the constrained h_{0i} control vertex is nonzero, so the probe is not vacuous), leaving only instantaneous constrained mixing that cannot form a cone. What *is* generated is stated with coefficients: a vacuum-energy shift $m^4/(16\pi^2\varepsilon)$ of the source ρ_m through the same MI form factor (the law — a_0, s, K — untouched), a longitudinal W^2 operator starting at $O(\delta u^4)$, and a curvature-cross term $-(s/6)m^2RW$ at $2H^2/m^2\sim3\times10^{-84}$ of the leading term for proton-mass matter. Dressed Källén–Lehmann positivity holds everywhere under the real KMS convolution of the exact cut density (branch points resolved; detailed balance to 4×10^{-12}; a hand-injected O(1) ghost *is* flagged, and the detector's blind spot — integrated negative weight $\lesssim0.15$ — is measured and stated: positivity is necessary, not sufficient). Nothing flips between the two a_0 footings. **Still open at one loop (named, not walled):** the $T_{\mu\nu}$/disformal coupling variant of ρ_m (argued to give the same conclusions since $W_{dS}=0$, not computed); the one-loop *finite* nonlocal parts and light-field IR on dS; the all-n proof of the TT-vertex zero (CAS-proven n=1,2; derivative-counting beyond); pure-graviton diagrams beyond selection-rule + constraint-closure coverage; and everything two-loop. Two finite, unwalled *classical* computations are the natural next step: the full closed-form nonlocal stress tensor $T_{\mu\nu}=-(2/\sqrt{-g})\delta S_{matter}/\delta g$ with machine-verified on-shell conservation, and the coupled nonlinear Cauchy problem for the branch-cut class built on it. A first-principles ghost-condensate profile law (§5) also remains open.

5. Gravitational lensing: a disformal photon metric (universal, Cassini-safe, no double-counting)
In modified inertia the dynamics come from the inertial response, not the metric: the framework reproduces the RAR $g\_obs=\sqrt{(g\_bar^2+g\_bar\ a_0)}$ using the *baryonic* field g_bar. A single-metric audit (`mi_lensing_doublecount_audit.py`) draws the sharp consequence. With one metric, light — having no inertia — sees that baryonic metric and *under*-lenses by ν; and trying to fix this by enhancing the metric **double-counts**, because the same modified-inertia response then over-predicts the rotation curve by the enhancement factor (consistency forces the metric back to baryonic). Full lensing in a single metric would demand zero inertia modification — i.e. pure modified gravity, which reinstates the Cassini quadrupole. So the enhancement cannot be shared between one metric and the inertia; this is why the earlier drafts fell back on a hand-placed condensate.

The resolution is a **disformal photon metric** built from the passive frame (`mi_disformal_lensing.py`): light propagates on $\tilde{g}\_\mu\nu = g\_\mu\nu + B\ u\_\mu u\_\nu$, while matter keeps modified inertia in g. The conformal part leaves null rays untouched, so only the disformal $u\_\mu u\_\nu$ term bends light; setting the resulting lensing potential $(\Phi+\Psi)/2 = \Phi-B/4$ equal to the MOND potential fixes $B = 4(\Phi\_bar-\Phi\_MOND) > 0$, tied to the *same* ν the RAR already uses. Three properties are computed. It is **universal** — one rule for every galaxy, not the per-galaxy placement a bolted-on condensate requires. It is **double-count-free by construction** — the matter dynamics use g, untouched; only light uses $\tilde{g}$. And it is **Cassini-safe**: $B\sim(\nu-1)\longrightarrow a_0/2a\longrightarrow 0$ in the solar system, so $\tilde{g}\longrightarrow g$ and light bending reduces to GR, with a PPN shift $|\Delta\gamma|\sim 10^{-13}-10^{-11}$ at the Sun — six to nine orders under the 2.3×10^{-5} Cassini bound — and no dynamical-aether bill, the frame being passive. In the weak field $B\sim 3\times10^{-7}\ll 1$, so $\tilde{g}$ is a valid Lorentzian metric. The two-metric coupling is **ghost-free and causal at the kinematic level** (`mi_disformal_ghostfree.py`): $\tilde{g}$ is Lorentzian and invertible for B<1 (physical $B\sim 3\times10^{-7}$); photons are **subluminal** relative to g (since B>0, the $\tilde{g}$-null wave-covector is g-spacelike, inside the g light-cone — no closed timelike curves; a B<0 would be acausal, so lensing and causality pick the same sign); the photon is ordinary Maxwell on a Lorentzian background — two healthy polarizations, no ghost; and B, being an *algebraic* function of the local acceleration like the passive frame itself, adds **no new propagating degree of freedom**, leaving the §4 constraint structure (0 frame dof) intact — total propagating content 2 (graviton) + 2 (photon). The full *nonlinear coupled* Ostrogradsky check is now **done** (`mi_disformal_ostrogradsky.py`): because the frame is passive (§4, 0 dof) the acceleration is $a\sim\nabla\Phi$, first-order in the metric, so the disformal Lagrangian depends only on *first* derivatives of every dynamical field (∂A and ∂g) — Ostrogradsky's second-derivative hypothesis is never met, hence no higher-derivative ghost at any order (a machine check confirms the graviton fluctuation enters at most at first order, ∂h, never $\partial^2 h$). The one hinge is §4 passivity itself: a *dynamical* khronon would give $a\sim\partial^2 T$ and reintroduce the concern, but §4 forbids it. **Resolved (`mi_disformal_locality.py`):** the enhancement field $g\_obs=\nu(g\_bar)g\_bar$ is curl-free — so a local $B(a/a_0)$ is an exact potential — for spherical symmetry (curl computed to be identically 0), but has order-unity curl for non-spherical mass distributions, so a purely local B is only the spherical approximation. In general B is **non-local**, and the required non-locality is precisely the framework's own K(□_u): an AQUAL/Poisson lensing potential $\nabla\cdot[\mu(|\nabla\Phi\_M|/a_0)\nabla\Phi\_M]=\nabla\cdot g\_bar$, curl-free by construction and momentum-conserving for any geometry, reducing to the local B in spherical symmetry. So B is fixed by the *same* non-locality that gives the dynamics — **not a new ingredient**. The only residual is the explicit K(□_u) lensing-potential solution for a specific non-spherical galaxy (a standard AQUAL-type computation, no new physics) and the exact closed-form B(ν). This is the honest route to dark-matter-free lensing — universal, Cassini-safe, double-count-free, and ghost-free at the kinematic level — superseding the fine-tuned ghost-condensate. The genuinely distinctive modified-inertia signature lives in the *dynamical* sector (the MG-impossible non-adiabatic velocity-dispersion hysteresis), not in lensing.

6. Honest standing

Reasoning from the framework's own passive-frame, MOND-in-matter premise: the worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity (at the frame-equation and principal-symbol level); the mixed two-point propagator is causal, ghost-free (Källén–Lehmann), and principal-symbol-well-posed; the Einstein-aether strong-coupling wall is a misapplication that does not bind the framework (it binds only the rejected modified-gravity limb); and — new in v4 — the passive-frame **constraint structure closes as a machine-verified Dirac system with zero propagating frame modes**, the nonlocal operator **K(□_u) is rigorously defined** by a unique positive Herglotz–Nevanlinna measure, and **nonlinear classical stability holds** (the no-wave-cone symbol to all orders). Gravitational lensing is resolved by a disformal photon metric $\tilde{g}=g+B\,u_\mu u_\nu$ (light-only, B tied to ν): universal, double-count-free, and Cassini-safe ($|\Delta\gamma|\sim10^{-13}$ at the Sun, $\ll 2.3\times10^{-5}$) — superseding the fine-tuned condensate; ghost-free and causal *nonlinearly* ($\tilde{g}$ Lorentzian, photons subluminal, no new dof; the coupled Ostrogradsky check is done — a first-order Lagrangian given §4 passivity), the one hinge being the passive frame. The completion is therefore **not** the modified-gravity limb (Cassini-failing) and **not yet** a finished field theory — it is a well-posed worldline theory whose passive-frame covariantization has a closed constraint algebra (0 frame dof), a rigorously defined causal-retarded nonlocal operator, a causal ghost-free two-point sector, principal-symbol well-posedness, and nonlinear stability, with three honest open edges: the full closed-form metric stress tensor $T_{\mu\nu}=-(2/\sqrt{-g})\delta S_{matter}/\delta g$ with on-shell conservation (and the coupled nonlinear Cauchy problem built on it), and the one-loop residuals now named precisely (the disformal ρ_m variant, finite nonlocal parts, the all-n TT-vertex proof, pure-graviton diagrams, two loops) — the one-loop *divergence* structure itself is computed in v11: a_0 unrenormalized, no aether kinetic term or wave cone generated, the graviton-frame mixing channel closed, dressed positivity preserved, all under a stated matter-density proxy. (The disformal photon metric is now ghost-free and causal *nonlinearly* — $\tilde{g}$ Lorentzian, photons subluminal, no new dof, and the coupled Ostrogradsky check done as a first-order Lagrangian given §4 passivity; the disformal B is fixed by the framework's own nonlocal K(□_u) — an AQUAL-type lensing potential, momentum-conserving for any geometry (local B is its exact spherical limit), so it is not a new ingredient; the only residual is a standard AQUAL-type solve for a specific non-spherical galaxy.) The MOND sign stays postulated and a_0's value underived. No completeness or theory-of-everything claim is made.

7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet–Skordis khronon [2025], the nonlocal mimetic model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame sector is Einstein-aether [Jacobson & Mattingly 2001], its PPN structure [Foster & Jacobson 2006], its GW170817 constraint [Gong et al. 2018], and its strong-coupling corner [Blas, Pujolàs & Sibiryakov 2009]; the nonlocal well-posedness class is Källén–Lehmann/Herglotz–Nevanlinna (passive-causal), distinct from the entire-function class [Biswas et al.; Barnaby & Kamran]. The doubled-worldline formalism is Galley [2013]; the lensing constraint is Brouwer et al. [2021] and Mistele & McGaugh [2024].

8. Reproducibility

Committed scripts (repository `real_research/reviews/`): the worldline sector and Cassini evasion (`cassini_mi_evasion_2026/`); the covariant kinetic wins (`mi_kinetic_completion_2026/`); the mixed-propagator causality + Källén–Lehmann positivity (`mi_propagator_2026/`); the framework-first correction of the strong-coupling wall (`mi_strongcoupling_framework_2026/`); the all-orders Cauchy checks (`mi_cauchy_wellposed_2026/`); the lensing fine-tuning analysis (`mi_lensing_2026/`); and — new in v4 — the closed Dirac constraint analysis, the no-wave-cone crux symbol, the Herglotz–Nevanlinna operator definition, and the nonlinear/radiative stability analysis, and the graviton principal-symbol block-diagonality via the K⟶1 UV limit (`mi_formal_completion_2026/`: `A10_dirac_block.py`, `A6b_higher_n.py`, `constraint_structure.py`, `crux_variation_check.py`, `operator_definition.py`, `stability.py`, `transverse_mode_analysis.py`, `principal_symbol_blockdiag.py`, `mi_lensing_from_stress_tensor.py`, `mi_lensing_doublecount_audit.py`, `mi_disformal_lensing.py`, `mi_disformal_ghostfree.py`, `mi_disformal_ostrogradsky.py`, `mi_disformal_locality.py`; and — new in v11 — the one-loop dS radiative computation: `mi_oneloop_desitter.py` (capstone; every check earned, superseding a v1 draft whose decisive check was hard-coded), `oneloop_laneA_divergences.py`, `oneloop_laneB_mixing.py`, `oneloop_laneC_positivity.py`; and — new in v12 — the finite parts and two-loop matter sector: `twoloop_laneA_finite.py` (bounded-below + dS IR gap), `twoloop_laneC_a0.py` (a_0 additive non-renormalization to all orders), `twoloop_aether_resolution.py` (the corrected two-loop aether-term analysis: F1 + K(0)=0, and the artifact it replaces); the ordering-invariance check (`twoloop_ordering_fork.py`: the □_u ordering is immaterial by unit-norm); and — new in v13 — the two-loop graviton-sector census (`twoloop_graviton_seagull_vertex.py`, `twoloop_graviton_TTloop.py`, `twoloop_graviton_cone_vs_mass_decider.py`, `twoloop_graviton_kperp_rationing_alln.py`, `twoloop_graviton_qfree_diagnosis.py`)). Sign-wall inputs: `ACTIVE_KERNEL_SIGNTHEOREM_2026-06.md`, `FOURTH_ORDER_SIGN_WALL_2026-07.md`; trichotomy: `COVARIANT_MI_COMPLETION_2026-06.md`.

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14. Companion (this program): sign wall, Zenodo 10.5281/zenodo.21184373; covariant-MI quadrichotomy, 10.5281/zenodo.21148494; written MI action v1–v6 and v7–v10, 10.5281/zenodo.21253645 / 21260830 / 21263846 / 21264727 / 21265804 / 21266571 / 21266966 / 21267105 / 21267312 / 21267493.

*Both a_0 footings throughout. Each claim labeled derived / postulated / open. New in v4–v5: the passive-frame constraint structure closes as a machine-verified Dirac system (0 propagating frame dof), the nonlocal operator K(□_u) is rigorously defined by a unique positive Herglotz–Nevanlinna measure, and nonlinear classical stability holds — the graviton-cone step now computed via the K⟶1 UV limit. The "MOND does not force modified gravity" claim is established at the frame-equation and principal-symbol level; the full metric stress tensor T_μν and the coupled nonlinear Cauchy problem are named open classical computations alongside the covariant one-loop de Sitter edge. New in v7: gravitational lensing is resolved by a disformal photon metric $\tilde{g}=g+B\,u_\mu u_\nu$ (light-only, B tied to the same ν as the RAR) — universal (one rule, all galaxies), double-count-free by construction, and Cassini-safe ($|\Delta\gamma|\sim10^{-13}$ at the Sun, $\ll 2.3\times10^{-5}$), superseding the fine-tuned ghost-condensate; the single-metric route was shown obstructed by double-counting. New in v8–v9: the two-metric coupling is ghost-free and causal *nonlinearly* ($\tilde{g}$ Lorentzian, photons subluminal vs g, no new propagating dof, §4 constraints intact) — the coupled Ostrogradsky check is done, the disformal Lagrangian being first-order in every dynamical field given §4 passivity (the one hinge). New in v10: the disformal function B is fixed by the framework's own nonlocal K(□*u) (an AQUAL-type, momentum-conserving lensing potential; local B is its exact spherical limit) — not a new ingredient. New in v11: the one-loop de Sitter radiative structure is computed at the divergence level — a_0 unrenormalized (exact measure, unit resolvent weight), no transverse aether kinetic term or wave cone generated (graviton-frame mixing closed by exact curvature-commutator algebra; TT×frame vertex exactly zero), dressed Källén–Lehmann positivity and KMS detailed balance preserved — under a stated $\rho\_m=m^2\varphi^2$ proxy, with the disformal variant, finite parts, and two loops named open. New in v12: the one-loop finite parts are bounded-below and dS-IR-regulated (friction gap 3H/2), a_0's additive non-renormalization is all-orders exact (shift symmetry + unit-norm), and the two-loop matter-sector transverse aether term is not generated (F1 geodesy + K(0)=0; a naive matter-bubble 'generation' shown to be a mis-assigned-vertex artifact) — the graviton-loop sector is also closed at divergence level (q*⊥ rationed, TT vertex vanishing all-n, h_{0i} instantaneous) — the passive frame survives two loops at divergence level in both sectors; finite parts, higher loops, and the disformal variant remain open. The MOND sign is a postulate; the Einstein-aether wall is a mis-application. No completeness or theory-of-everything claim.*